

Algebra II Sequences and Series Cheatsheet

Module 8

Sequence vs Series

sequence: list of terms

series: sum of terms

Arithmetic Sequences

$$a_n = a_1 + (n - 1)d$$

$$d = a_n - a_{n-1}$$

Recursive form:

$$a_n = a_{n-1} + d, \quad a_1 = \text{first term}$$

Arithmetic Series

$$S_n = \frac{n(a_1 + a_n)}{2}$$

$$S_n = \frac{n}{2} (2a_1 + (n - 1)d)$$

Geometric Sequences

$$a_n = a_1 r^{n-1}$$

$$r = \frac{a_n}{a_{n-1}}$$

Recursive form:

$$a_n = r a_{n-1}, \quad a_1 = \text{first term}$$

Geometric Series

$$S_n = \frac{a_1(1 - r^n)}{1 - r} \quad (r \neq 1)$$

Infinite geometric series:

$$S_\infty = \frac{a_1}{1 - r} \quad (|r| < 1)$$

If $|r| \geq 1$, the infinite geometric series diverges.

Sigma Notation

$$\sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

$$\sum_{k=1}^n c = cn$$

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

Quick Checks

- Arithmetic means constant difference.
- Geometric means constant ratio.
- Infinite geometric sums require $|r| < 1$.
- Match the starting index to the first term.